



Qualification Pack

Transformer Manufacturing Supervisor

QP Code: CUTM/PSS/Q0102

Version: 1.0

NSQF Level: 5

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CUTM/PSS/Q0102: Transformer Manufacturing Supervisor

Brief Job Description

The Transformer Manufacturing Supervisor course trains individuals in supervising transformer production, quality control, and assembly processes. It covers manufacturing techniques, material selection, safety protocols, and compliance with industry standards. The course equips learners with technical expertise, leadership skills, and process optimization knowledge to ensure efficient transformer manufacturing and workplace safety.

Personal Attributes

A Transformer Manufacturing Supervisor must possess technical expertise in transformer production, leadership skills to manage teams, and problem-solving abilities to address manufacturing challenges. Strong attention to detail, communication, and organizational skills are essential. They should also ensure quality compliance, safety adherence, and efficient workflow management to optimize transformer manufacturing processes.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/PSS/N0103: Understand transformer manufacturing requirements](#)
2. [CUTM/PSS/N0104: Perform transformer manufacturing process.](#)
3. [CUTM/VSQ/N7801: Ensure organizational health, safety, and security](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Power Distribution
Occupation	Transformer Manufacturing
Country	India
NSQF Level	5
Credits	17

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Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	Pursuing 2nd year of UG (B.E./B. Tech in Mechanical / Electrical / Electronics / Production / Mechatronics Engineering and continuing education) with NA of experience OR Completed 2nd year of UG (UG Diploma) (B.E./B. Tech in Mechanical / Electrical / Electronics / Production / Mechatronics Engineering) with NA of experience OR Completed 3-year diploma (after 10th) (Mechanical / Electrical / Electronics / Production / Mechatronics Engineering) with 1 Year of experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	22/06/2026
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	QG-05-PW-00615-2023-V1-CUTM
NQR Version	1.0

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CUTM/PSS/N0103: Understand transformer manufacturing requirements

Description

The Individual is responsible to understand the design specification, verification of procured material, and manufacturing transformer for quality transformer fabrication along with following necessary safety norms.

Scope

The scope covers the following :

- The NOS "Understand Transformer Manufacturing Requirements" includes analyzing customer specifications, interpreting engineering drawings, identifying required tools and materials, understanding the bill of materials (BoM), ensuring compliance with manufacturing standards, maintaining documentation, and coordinating with procurement and production teams to facilitate efficient transformer fabrication while adhering to quality and safety norms.

Elements and Performance Criteria

Understanding the customer requirement, engineering drawing and standards.

To be competent, the user/individual on the job must be able to:

- PC1.** interact with the head of transformer manufacturing on customer requirement, engineering drawings and standards to be followed for transformer manufacturing process.
- PC2.** understand tools, raw materials and consumables required for production in sufficient quantity as per production plan/operators' instructions.
- PC3.** clearly understand the application of transformer and standards to be followed for manufacturing and testing process.

Reading of detailed assembly drawing, dimensional tolerances and assembly process.

To be competent, the user/individual on the job must be able to:

- PC4.** understand detailed assembly drawing, material grade, dimensional tolerances, and assembly process for the specific order to be executed.
- PC5.** distribute the assembly drawing, standards, and process steps to be followed for each stage.
- PC6.** distribute the work order details required for each process, batch details to be followed for the manufacturing stage.
- PC7.** get necessary clarification from the appropriate authorities in case of any doubts.

Study of bill of materials.

To be competent, the user/individual on the job must be able to:

- PC8.** understand bill of materials along with quantity required transformer manufacturing process with the expected quality requirements.
- PC9.** ensure that required raw material/tools is procured from stores /vendor before starting work.
- PC10.** check incoming quality Inspection reports on meeting the quality and quantity requirements.
- PC11.** ensure tools required are placed in the respective stages of transformer manufacturing process.
- PC12.** if tool is not available collect the tools from stores / vendor site and place it in the respective stages.

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- PC13.** update all the material transaction details in appropriate logs, template, MIS.
- PC14.** maintain production file, log, template MIS and update as per requirement.
- PC15.** report to Head of manufacturing in case of any dispute in the drawing and standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** customer requirements, engineering drawings, and applicable transformer manufacturing standards.
- KU2.** Identify tools, raw materials, and consumables required for transformer production.
- KU3.** transformer components, including core, coil, insulation, and cooling systems.
- KU4.** material grades, dimensional tolerances, and assembly processes for transformer manufacturing.
- KU5.** different transformer applications and their impact on design and manufacturing.
- KU6.** bill of materials (BoM) and procurement processes to ensure availability of required materials.
- KU7.** quality control procedures, inspection methods, and acceptance criteria for transformers.
- KU8.** safety regulations, workplace hazards, and best practices for maintaining a safe working environment.
- KU9.** documentation, production logs, and reporting requirements for compliance and traceability.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret engineering drawings, standards, and customer specifications accurately.
- GS2.** Communicate effectively with team members, supervisors, and procurement personnel regarding material and process requirements.
- GS3.** Document and maintain records such as bill of materials (BoM), production logs, and quality reports for compliance and traceability.
- GS4.** Plan and organize transformer manufacturing tasks, ensuring proper sequencing of activities for efficient workflow.
- GS5.** Analyze and evaluate raw material requirements, ensuring the availability of required components before starting production.
- GS6.** Apply problem-solving skills to address discrepancies in material specifications, design requirements, or assembly processes.
- GS7.** Make informed decisions regarding process adjustments based on transformer design and manufacturing feasibility.
- GS8.** Ensure customer-centricity by aligning production processes with client requirements and quality expectations.
- GS9.** Think critically and analytically to identify potential process improvements and optimize manufacturing efficiency.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding the customer requirement, engineering drawing and standards.</i>	6	18	-	-
PC1. interact with the head of transformer manufacturing on customer requirement, engineering drawings and standards to be followed for transformer manufacturing process.	2	6	-	-
PC2. understand tools, raw materials and consumables required for production in sufficient quantity as per production plan/operators' instructions.	2	6	-	-
PC3. clearly understand the application of transformer and standards to be followed for manufacturing and testing process.	2	6	-	-
<i>Reading of detailed assembly drawing, dimensional tolerances and assembly process.</i>	8	13	-	-
PC4. understand detailed assembly drawing, material grade, dimensional tolerances, and assembly process for the specific order to be executed.	2	5	-	-
PC5. distribute the assembly drawing, standards, and process steps to be followed for each stage.	2	2	-	-
PC6. distribute the work order details required for each process, batch details to be followed for the manufacturing stage.	2	2	-	-
PC7. get necessary clarification from the appropriate authorities in case of any doubts.	2	4	-	-
<i>Study of bill of materials.</i>	16	39	-	-
PC8. understand bill of materials along with quantity required transformer manufacturing process with the expected quality requirements.	2	3	-	-
PC9. ensure that required raw material/tools is procured from stores /vendor before starting work.	2	6	-	-
PC10. check incoming quality Inspection reports on meeting the quality and quantity requirements.	2	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure tools required are placed in the respective stages of transformer manufacturing process.	2	6	-	-
PC12. if tool is not available collect the tools from stores / vendor site and place it in the respective stages.	2	3	-	-
PC13. update all the material transaction details in appropriate logs, template, MIS.	2	6	-	-
PC14. maintain production file, log, template MIS and update as per requirement.	2	3	-	-
PC15. report to Head of manufacturing in case of any dispute in the drawing and standards.	2	6	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0103
NOS Name	Understand transformer manufacturing requirements
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQC Clearance Date	23/06/2023

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CUTM/PSS/N0104: Perform transformer manufacturing process.

Description

The NOS "Perform Transformer Manufacturing Process" covers executing core assembly, coil winding, insulation, vapor phase drying, tank fabrication, and final transformer assembly. It includes adhering to design specifications, ensuring quality checks, following safety protocols, maintaining documentation, and coordinating with production teams to achieve efficient and standardized transformer manufacturing.

Scope

The scope covers the following :

- The NOS "Perform Transformer Manufacturing Process" includes executing core and coil assembly, insulation, vapor phase drying, oil impregnation, and tank fabrication. It covers ensuring material availability, maintaining quality standards, following safety protocols, conducting testing procedures, documenting production activities, and coordinating with teams to achieve efficient and high-quality transformer manufacturing.

Elements and Performance Criteria

Confirm availability of required quantity as per transformer design.

To be competent, the user/individual on the job must be able to:

- PC1.** check quantity available at each process stage as per transformer design.
- PC2.** report the concern in case of any quality or quantity discrepancy.
- PC3.** arrange for discrepant quantity or quality of materials

Core Assembly

To be competent, the user/individual on the job must be able to:

- PC4.** understand core size, material, and grain orientation.
- PC5.** cut core shape as per design.
- PC6.** understand different core stacking methods.
- PC7.** use of temporary bolt guides for stacking.
- PC8.** stake core as per assembly drawing requirement with appropriate quantity.
- PC9.** tighten bolt to avoid gap between cores.
- PC10.** insulate assembled core.

Coil Assembly

To be competent, the user/individual on the job must be able to:

- PC11.** understand coil size, material and number of coils as per design.
- PC12.** keep winding room separate from other manufacturing areas with positive pressure and stake horizontal/vertical winding mandrels.
- PC13.** check conductor is either copper magnetic wire or continuously transposed conductor.
- PC14.** check conductor purchased are pre-wrapped with thermally upgraded paper or nomex.
- PC15.** check high strength wire or epoxy bonded CTC used when high short circuit forces required as per design requirement.

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- PC16.** chose winding type according to voltage & application, cont. disc and design interleaved disc, helical, etc.
- PC17.** make coil windings as per assembly requirement.
- PC18.** complete lead connection with stress ring and pick tail connected with lead.
- PC19.** complete winding insulation package.
- PC20.** complete lead braising.

Coil & Core Assembly

To be competent, the user/individual on the job must be able to:

- PC21.** bond exposed edges of core with low viscosity, high strength epoxy resin which penetrates and bonds laminations.
- PC22.** clamp prefabricated coil with insulations placed on bottom clamps.
- PC23.** clamp windings using external or internal tie rods to provide additional support for axial forces.
- PC24.** rigidly support leads and busbars to withstand forces from shipping and short circuits.
- PC25.** move assembly on air cushion.
- PC26.** make necessary LTC / DTC lead connection.
- PC27.** link board with re-connectable LV.

Vapor phase unit

To be competent, the user/individual on the job must be able to:

- PC28.** complete core and coil assembly is dried using a vapor phase cycle method.
- PC29.** maintain required oven temperature and time for curing.
- PC30.** continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.

Oil Tank fabrication

To be competent, the user/individual on the job must be able to:

- PC31.** understand tank requirements and study tank assembly drawing.
- PC32.** fabricate tank with appropriate process steps and materials.
- PC33.** maintain required oven temperature and time for curing.
- PC34.** continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.

Transformer Assembly

To be competent, the user/individual on the job must be able to:

- PC35.** understand tank requirements and study tank assembly drawing.
- PC36.** fabricate tank with appropriate process steps and materials.
- PC37.** make grit-blasted cleaning the tank before priming and painting.
- PC38.** provide necessary LTC connections.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** transformer core assembly, including material selection, cutting, stacking, and insulation techniques.
- KU2.** coil winding processes, including types of windings, conductor selection, and insulation methods.
- KU3.** coil and core assembly, ensuring proper alignment, clamping, and securing techniques.
- KU4.** vapor phase drying, oil impregnation, and insulation curing to enhance transformer performance.
- KU5.** fabrication and assembly of transformer tanks, including welding, surface treatment, and painting.
- KU6.** electrical and mechanical testing methods to verify transformer quality and compliance with standards.
- KU7.** safety protocols, hazard identification, and best practices in transformer manufacturing.
- KU8.** production planning, material management, and workflow optimization for efficient manufacturing.
- KU9.** documentation requirements, including bill of materials (BoM), production logs, and quality records.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret transformer manufacturing drawings, process instructions, and quality standards.
- GS2.** Communicate effectively with team members, supervisors, and engineers regarding production issues and solutions.
- GS3.** Document and maintain production records, quality checklists, and material usage logs for compliance and traceability.
- GS4.** Plan and organize transformer manufacturing processes, ensuring smooth execution of core assembly, coil winding, and insulation.
- GS5.** Identify and troubleshoot common manufacturing defects such as misalignment, insulation failure, or coil damage.
- GS6.** Make informed decisions regarding material selection, process adjustments, and quality assurance based on standards.
- GS7.** Ensure customer-centricity by maintaining precision in transformer assembly to meet client specifications.
- GS8.** Apply problem-solving skills to resolve issues related to machinery, raw materials, and assembly constraints.
- GS9.** Think critically and analytically to optimize production efficiency and reduce wastage.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Confirm availability of required quantity as per transformer design.</i>	6	9	-	-
PC1. check quantity available at each process stage as per transformer design.	2	3	-	-
PC2. report the concern in case of any quality or quantity discrepancy.	2	3	-	-
PC3. arrange for discrepant quantity or quality of materials	2	3	-	-
<i>Core Assembly</i>	14	19	-	-
PC4. understand core size, material, and grain orientation.	2	3	-	-
PC5. cut core shape as per design.	2	3	-	-
PC6. understand different core stacking methods.	2	3	-	-
PC7. use of temporary bolt guides for stacking.	2	3	-	-
PC8. stake core as per assembly drawing requirement with appropriate quantity.	2	3	-	-
PC9. tighten bolt to avoid gap between cores.	2	1	-	-
PC10. insulate assembled core.	2	3	-	-
<i>Coil Assembly</i>	20	32	-	-
PC11. understand coil size, material and number of coils as per design.	2	3	-	-
PC12. keep winding room separate from other manufacturing areas with positive pressure and stake horizontal/vertical winding mandrels.	2	3	-	-
PC13. check conductor is either copper magnetic wire or continuously transposed conductor.	1	2	-	-
PC14. check conductor purchased are pre-wrapped with thermally upgraded paper or nomex.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. check high strength wire or epoxy bonded CTC used when high short circuit forces required as per design requirement.	2	3	-	-
PC16. chose winding type according to voltage & application, cont. disc and design interleaved disc, helical, etc.	2	3	-	-
PC17. make coil windings as per assembly requirement.	3	6	-	-
PC18. complete lead connection with stress ring and pick tail connected with lead.	2	3	-	-
PC19. complete winding insulation package.	2	3	-	-
PC20. complete lead braising.	2	3	-	-
<i>Coil & Core Assembly</i>	15	24	-	-
PC21. bond exposed edges of core with low viscosity, high strength epoxy resin which penetrates and bonds laminations.	2	3	-	-
PC22. clamp prefabricated coil with insulations placed on bottom clamps.	2	3	-	-
PC23. clamp windings using external or internal tie rods to provide additional support for axial forces.	3	6	-	-
PC24. rigidly support leads and busbars to withstand forces from shipping and short circuits.	2	3	-	-
PC25. move assembly on air cushion.	2	3	-	-
PC26. make necessary LTC / DTC lead connection.	2	3	-	-
PC27. link board with re-connectable LV.	2	3	-	-
<i>Vapor phase unit</i>	8	12	-	-
PC28. complete core and coil assembly is dried using a vapor phase cycle method.	2	3	-	-
PC29. maintain required oven temperature and time for curing.	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC30. continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.	4	6	-	-
<i>Oil Tank fabrication</i>	10	15	-	-
PC31. understand tank requirements and study tank assembly drawing.	2	3	-	-
PC32. fabricate tank with appropriate process steps and materials.	2	3	-	-
PC33. maintain required oven temperature and time for curing.	4	6	-	-
PC34. continually monitor power factor & water extraction to make sure kerosene is vaporized and drawn by vacuum into autoclave.	2	3	-	-
<i>Transformer Assembly</i>	7	9	-	-
PC35. understand tank requirements and study tank assembly drawing.	2	3	-	-
PC36. fabricate tank with appropriate process steps and materials.	2	3	-	-
PC37. make grit-blasted cleaning the tank before priming and painting.	2	1	-	-
PC38. provide necessary LTC connections.	1	2	-	-
NOS Total	80	120	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/PSS/N0104
NOS Name	Perform transformer manufacturing process.
Sector	Power
Sub-Sector	
Occupation	Transformer Manufacturing
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	22/06/2026
NSQC Clearance Date	23/06/2023

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CUTM/VSQ/N7801: Ensure organizational health, safety, and security

Description

This NOS focuses on maintaining workplace safety, health, and security by identifying hazards, following safety procedures, using personal protective equipment (PPE), and responding to emergencies. It ensures compliance with health and safety regulations, proper waste disposal, and adherence to security protocols to create a safe and productive work environment.

Scope

The scope covers the following :

- This NOS covers workplace hazard identification, adherence to safety regulations, proper use of personal protective equipment (PPE), emergency response, fire safety, first aid, and waste management. It also includes implementing security measures, ensuring workplace hygiene, and promoting a safety-conscious culture to prevent accidents and maintain a secure working environment.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.
- PC2.** recognize other possible security issues existing in the workplace.
- PC3.** plan safety techniques for self and others.
- PC4.** recognize different measures to curb the hazards.
- PC5.** implement programs for health, safety and security of the people at the workplace.
- PC6.** communicate the safety plans to everyone at the workplace.
- PC7.** attach disciplinary rules with the implementation at the workplace.
- PC8.** comply with regulatory health and safety guidelines applicable to the workplace.
- PC9.** use and maintain protective equipment as per requirement in a production house.
- PC10.** ensure organizational activities in line with approved guidelines and procedures.
- PC11.** undertake and advise healthy lifestyle for self and others.
- PC12.** follow environment management system related procedures.
- PC13.** ensure safe handling and disposal of waste and debris.
- PC14.** minimize health and safety risks to self and others due to own actions.
- PC15.** monitor the workplace and work processes for potential risks and threats to self and others.
- PC16.** carry out periodic check to keep work area free from hazards and obstructions.
- PC17.** ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.
- PC18.** organize first aid, firefighting and emergency training sessions for others.
- PC19.** instruct and guide others to take action in the event of fire, emergencies or accidents.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines, workplace policies, and compliance with national and industry safety standards.
- KU2.** common hazards such as electrical risks, fire, chemical exposure, and physical injuries.
- KU3.** different types of PPE (gloves, helmets, safety shoes, goggles) and their correct usage.
- KU4.** the proper handling, operation, and maintenance of workplace tools to prevent accidents.
- KU5.** Awareness of first aid procedures, fire evacuation plans, and emergency contact protocols.
- KU6.** fire hazards, fire extinguisher types, and safe fire-handling procedures in case of an incident.
- KU7.** the importance of keeping the workplace clean, organized, and free of potential hazards.
- KU8.** proper lifting posture, use of lifting tools, and avoiding strain-related injuries.
- KU9.** workplace security protocols, restricted areas, and unauthorized access prevention.
- KU10.** safe disposal methods for hazardous and non-hazardous materials to ensure environmental compliance.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** report safety concerns, follow instructions, and communicate hazards to supervisors and colleagues.
- GS2.** Identify workplace safety issues and taking appropriate actions to prevent accidents.
- GS3.** Handle safety-related measurements like voltage limits, weight capacity, and fire extinguisher ratings.
- GS4.** Ensure timely completion of tasks while following safety protocols and risk management procedures.
- GS5.** Work effectively with colleagues to maintain a safe and secure work environment.
- GS6.** Identify risks such as electrical shocks, fire, slippery surfaces, and chemical exposure in a workplace.
- GS7.** Ensure the safe use of machines, hand tools, and electrical equipment to prevent accidents.
- GS8.** Follow safety protocols, emergency procedures, and workplace safety policies.
- GS9.** do basic first aid techniques, fire safety procedures, and responding to workplace accidents.
- GS10.** Ensure proper disposal of hazardous and non-hazardous waste in compliance with environmental standards.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	70	-	-
PC1. identify different kinds of possible hazards at the workplace hazards: environmental, personal, ergonomic and chemical.	2	4	-	-
PC2. recognize other possible security issues existing in the workplace.	2	4	-	-
PC3. plan safety techniques for self and others.	2	4	-	-
PC4. recognize different measures to curb the hazards.	2	4	-	-
PC5. implement programs for health, safety and security of the people at the workplace.	2	4	-	-
PC6. communicate the safety plans to everyone at the workplace.	2	4	-	-
PC7. attach disciplinary rules with the implementation at the workplace.	2	4	-	-
PC8. comply with regulatory health and safety guidelines applicable to the workplace.	2	4	-	-
PC9. use and maintain protective equipment as per requirement in a production house.	2	3	-	-
PC10. ensure organizational activities in line with approved guidelines and procedures.	2	3	-	-
PC11. undertake and advise healthy lifestyle for self and others.	1	3	-	-
PC12. follow environment management system related procedures.	1	3	-	-
PC13. ensure safe handling and disposal of waste and debris.	2	4	-	-
PC14. minimize health and safety risks to self and others due to own actions.	1	4	-	-
PC15. monitor the workplace and work processes for potential risks and threats to self and others.	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. carry out periodic check to keep work area free from hazards and obstructions.	1	4	-	-
PC17. ensure participation of self and others in mock drills/ evacuation procedures organized at the workplace.	1	4	-	-
PC18. organize first aid, firefighting and emergency training sessions for others.	1	3	-	-
PC19. instruct and guide others to take action in the event of fire, emergencies or accidents.	1	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7801
NOS Name	Ensure organizational health, safety, and security
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Detailed Guidelines for Assessment – Transformer Testing Supervisor (NSQF Level 5)

Assessment Mode: Theory, Practical, and Viva-Based Evaluation

The assessment for this qualification will be conducted in two components:

Theory Examination (40%) – Written MCQs and descriptive questions.

Practical Assessment (60%) – Hands-on evaluation in a lab/testing environment.

1. General Assessment Strategy

- The assessment should take place in a controlled environment, such as a transformer testing lab or training facility.
- Batches will be assigned to Assessment Agencies, which will deploy ToA-certified Assessors to conduct the assessment.



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- The minimum passing percentage is 70% overall and 70% in each NOS/module.
- Surprise audits and video call verification will be conducted to ensure quality assurance.

2. Assessment for Each NOS

NOS 1: Understand Transformer Testing Requirements

Assessment Components:

- Theory Exam (30 marks) – Covers topics like testing standards, transformer specifications, tools, and raw materials.
- Practical Exam (70 marks) – Includes verification of test equipment, calibration checks, and documentation procedures.

NOS 2: Testing and Quality Check of Assembled Transformer

Assessment Components:

- Theory Exam (40 marks) – Includes transformer testing methods, inspection checklists, fault identification, and quality control.
- Practical Exam (60 marks) – Hands-on testing, measuring electrical parameters, and analyzing defects.

NOS 3: Ensure Organizational Health, Safety, and Security

Assessment Components:

- Theory Exam (30 marks) – Covers hazard identification, safety procedures, and emergency protocols.
- Practical Exam (70 marks) – Demonstration of PPE usage, risk mitigation, and workplace safety compliance.

NOS 4: Employability Skills

Assessment Components:

- Theory Exam (30 marks) – Includes career development, digital skills, communication, and financial literacy.
- Practical Exam (70 marks) – Role plays, case studies, and presentations on workplace ethics and teamwork.

3. Assessment Conditions

- Qualified Assessors: B.Tech in Electrical/Mechanical/Electronics with 3 years of industry experience and 1 year of training experience.
- Lead Assessors & Proctors: Must have at least 7 years of industry experience.
- Assessment Location: Conducted in a transformer testing facility or training center with access to

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equipment and testing tools.

- Assessment Tools & Equipment: Includes insulation testers, power analyzers, winding resistance meters, and digital multimeters.

- Mode of Assessment: Offline

4. Assessment Documentation:

- Geo-tagged photos of assessments.
- Video recordings for monitoring and auditing.
- Hard copies and soft copies of assessment records.

5. Passing Criteria & Reassessment

- Minimum Pass Percentage: 70% in aggregate and 70% in each NOS/module.
- Candidates failing the assessment will be given one reassessment opportunity after additional training.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/PSS/N0103.Understand transformer manufacturing requirements	30	70	-	-	100	30
CUTM/PSS/N0104.Perform transformer manufacturing process.	80	120	-	-	200	40
CUTM/VSQ/N7801.Ensure organizational health, safety, and security	30	70	0	0	100	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	160	290	0	0	450	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.